

From Static to Dynamic Detection

SIEM Rules Enhancement Report:
Gonalo Cunha – 108352
Miguel Vila – 107276

Summary

This report documents the significant improvements made to our SIEM rules implementation, transitioning from a static threshold-based approach (Dataset 8 analysis) to a dynamic, statistically-driven detection system (Dataset 10 analysis). The enhancements focus on improved accuracy, reduced false positives, and more granular threat identification.

1. Statistical Threshold Determination

Previous Approach: Fixed, arbitrary thresholds across all rules

- Protocol violations: Fixed 10% increase triggers
- Traffic anomalies: Static 100% thresholds
- Highly sensitive Geographic patterns (both by country and ASN)

Current Approach: Dynamic statistical analysis

- Baseline Establishment: Using 5th-95th percentiles for normal ranges
- Standard Deviation Analysis: DNS C&C detection now uses statistical outliers (305.2 std above mean)
- Coefficient of Variation: External access patterns analyzed using CV values (0.234 indicating robotic precision)

2. Rule Modularization and Specialization

Previous Approach: Monolithic client behavior with many analyses within the same SIEM rule.

Current Approach: Six specialized SIEM rules with focused objectives:

- Rule 1: Internal Botnet Detection (volume-based + lateral movement)
- Rule 2: HTTPS Data Exfiltration (upload/download ratio analysis)
- Rule 3: DNS Data Exfiltration (HTTPS/DNS ratio analysis)
- Rule 4: DNS C&C Activities (statistical outlier detection)
- Rule 5: Anomalous External Destinations (geographic pattern analysis)
- Rule 6: External Access Anomalies (timing pattern analysis for servers)

3. Enhanced Threat Identification and Scoring

Previous Approach: Basic IP listing with limited context

Current Approach: Comprehensive threat assessment with explicit reasoning:

IP Address	Rules Triggered	Threat Score
192.168.110.191	Botnet Volume Crit. (4), DNS Exfil. Crit. (4), DNS C&C Crit. (4)	12
192.168.110.196	Botnet Lateral (4)	4
192.168.110.193	HTTPS Exfil. Cri. (4)	4

Previous Approach: Narrative-heavy descriptions with minimal tables

Current Approach: Structured tabular format for key findings:

- Network topology analysis with device counts
- Critical services identification with traffic volumes
- Geographic distribution with percentages and flow ratios
- Threat summary matrices with severity classifications

5. Enhanced Statistical Foundation

Previous Approach: Limited statistical backing for detection rules

Current Approach: Rigorous statistical analysis:

- Normal Range Calculations: [0.103, 0.114] for upload/download ratios
- Percentile-Based Thresholds: 95th percentile boundaries for outlier detection
- Correlation Analysis: Multi-vector threat identification across rules
- Risk Assessment: Geographic-based ASN categorization (90.7% from never-contacted countries)

6. Comprehensive Baseline Establishment

Previous Approach: Basic protocol and port analysis

Current Approach: Multi-dimensional baseline including:

- Protocol distribution (88.1% TCP, 11.9% UDP)
- Normal ratio ranges with statistical bounds